Using this information, estimate the following quantities (a) total amount of charge transferred between cloud and ground (b) the current in the lightning bolt (c) the power delivered in 0.2 s.

Ans: charge = 20 C, I = 100 A, P = 5 GW

3. A copper wire of 10^{-6} m^2 area of cross section, carries a current of 2 A. If the number of electrons per cubic meter is 8×10^{28}, calculate the current density and average drift velocity.

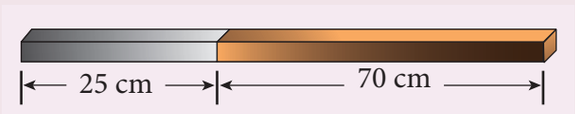
Ans: J = 2×10^6 Am^{-2} ; $v_d = 15.6 \times 10^{-5}$ ms^{-1}

4. The resistance of a nichrome wire at 0 ^{0}C is 10 Ω. If its temperature coefficient of resistance is 0.004/^{0}C, find its resistance at boiling point of water. Comment on the result.

Ans: R_T= 14 Ω.

As the temperature increases the resistance of the wire also increases.

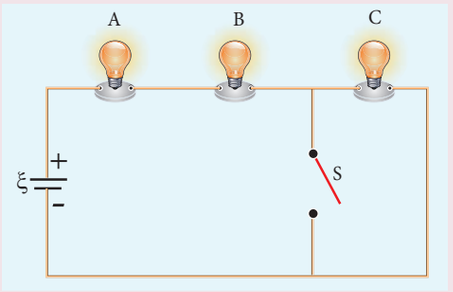
5. The rod given in the figure is made up of two different materials.



Both have square cross sections of 3 mm side. The resistivity of the first material is 4 x 10^{-3} Ω.m and it is 25 cm long while second material has resistivity of 5 x 10^{-3} Ω.m and is of 70 cm long. What is the resistivity of rod between its ends?

Ans: 500 Ω

6. Three identical lamps each having a resistance R are connected to the battery of emf as shown in the figure.



Suddenly the switch S is closed. (a) Calculate the current in the circuit when S is open and closed (b) What happens to the intensities of the bulbs A,B and C. (c) Calculate the voltage across the three bulbs when S is open and closed (d) Calculate the power delivered to the circuit when S is opened and closed (e) Does the power delivered to the circuit decreases, increases or remain same?

Ans:

Electrical quantities	Switch S is open	Switch S is closed
Current	$\frac{\xi}{3R}$	$\frac{\xi}{2R}$
Voltage	$V_A = \frac{\xi}{3R}$, $V_B = \frac{\xi}{3R}$, $V_C = \frac{\xi}{3R}$	$V_A = \frac{\xi}{2R}$, $V_B = \frac{\xi}{2R}$, $V_C = 0$
Power	$P_A = \frac{\xi^2}{9R}$, $P_B = \frac{\xi^2}{9R}$, $P_C = \frac{\xi^2}{9R}$	$P_A = \frac{\xi^2}{4R}$, $P_B = \frac{\xi^2}{4R}$, $P_C = 0$ Total power increases
Intensity	All the bulbs glow with equal intensity	The intensities of the bulbs A and B equally increase. Bulb C will not glow since no current pass through it.